Multiplication of algebraic fractions

1 Simplify the following expressions:

a $\frac{1}{a} \times \frac{1}{b}$

b $\frac{a}{3} \times \frac{a}{7}$

c $\frac{b}{q} \times \frac{k}{u}$

d $\frac{c}{3} \times \frac{d}{2}$

e $\frac{3y}{8} \times \frac{4y}{9}$

f $\frac{9x}{2} \times \frac{5y}{7}$

g $\frac{4x}{5} \times \frac{3y}{7}$

h $\frac{5u}{3a} \times \frac{3v}{5b}$

i $\frac{11s}{7t} \times \frac{3r}{5q}$

j $\frac{9a}{10b} \times \frac{3b}{8a}$

k $\frac{8x}{7y} \times \frac{9y}{5x}$

l $\frac{9u}{7v} \times \frac{8w}{11y}$

m $\frac{14u}{15v} \times \frac{40v}{24q}$

n $\frac{9u}{77v} \times \frac{110v}{24q}$.

o $\frac{x^2}{3} \times \frac{6}{x}$

p $\frac{p^2}{q} \times \frac{q^2}{p}$

q $\frac{r^2}{n} \times \frac{n^2}{r}$

r $\frac{5y^2}{9} \times 27y^2$

s $\frac{5y}{2} \times 6y^5$.

t $\frac{4m^2}{20} \times \frac{3y^2}{7m}$

u $\frac{7a^2}{15b^2} \times \frac{10b}{11a}$.

v $\frac{11a^2}{4b^2} \times \frac{10b}{3a}$

w $\left(-\frac{10}{21x}\right) \times \frac{3x^2}{110}$

x $\frac{2n^2}{6} \times \frac{-2y^2}{3n}$

2 Simplify:

a $\frac{m}{3} \times \frac{5m}{9} \times 81q$

b $\frac{2n}{3m} \times \frac{21q}{4n} \times \frac{10m}{49p}$

3 Simplify $\left(\frac{x}{3y}\right)^2 \times \frac{5x^2}{2y}$.

4 Complete the following:

$$\frac{\square}{3a^2} \times \frac{7b}{\square} = \frac{14bc}{9a^2d}$$

5 The product of two fractions is 1. If one of the fractions is $\frac{3x}{4yz}$, what is the other fraction?

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Division of algebraic fractions

6 Simplify the following expressions:

a $\frac{r}{3} \div \frac{2}{m}$

b $\frac{u}{3} \div \frac{4}{v}$

c $\frac{m}{4} \div \frac{m}{3}$

d $\frac{m}{28} \div \frac{23}{20}$

e $\frac{3y}{6} \div \frac{4y}{7}$

f $\frac{2x}{9} \div \frac{7}{5y}$

g $\frac{3y^2}{4} \div \frac{12}{29y^6}$

h $\frac{4u}{35y} \div \frac{14q}{50y}$

i $\frac{-2x}{11} \div \frac{2x}{3}$.

j $\frac{9u}{36v} \div \frac{7v}{36u}$

k $\frac{-9n}{4} \div \frac{11n}{8}$.

l $\frac{u^2}{12} \div \frac{u}{9}$

m $\frac{4x^2y}{9z^4} \div \frac{6xy^5}{15z}$

n $\frac{u^2}{9} \div \frac{u}{6}$.

o $\frac{8u^3}{32v} \div \frac{5v^3}{96u}$

p $\frac{3w^3}{7} \div 9w^6$

q $\frac{16u}{17v} \div (-14uv)$

r $\frac{45x^2yz}{10} \div \frac{9y^2}{5x}$

s $\frac{35t^2u^2}{12v^2w^2} \div \frac{35t^3u}{12w^2y^2}$

t $\frac{10t^2u^3}{35v^3w^3} \div \frac{10t^3u^2}{20w^3y}$

7 Simplify $\left(\frac{2x}{y}\right)^4 \div \left(\frac{z}{3x^2y}\right)^2$.

8 Simplify $\frac{3xy}{12y} \times \frac{4y^2}{3wx} \div \frac{15wx}{3w^2}$.

9 Complete the following:

$$\frac{\square}{4p^2} \div \frac{2q^2}{3r} = \frac{27r^3}{\square}$$

10 A rectangle has an area of $\frac{5x^3y^4}{3pq}$ and a length of $\frac{4xy}{p}$. Find an expression for the width of the rectangle.

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